

MAT2041 Assignment 3

The Chinese University of Hong Kong, Shenzhen
School of Data Science

Due: 17:00, March 13th, 2026

Instructions: Please ensure that you complete this assignment independently. If your solution appears to be identical or highly similar to others or generated using AI, you may be required to meet with the TA to verify your understanding. Cases of academic dishonesty may result in penalties, including a zero score for the assignment or other actions by the disciplinary committee. If you have any questions or need assistance, feel free to reach out to the course staff for guidance. We encourage you to engage in the learning process with integrity and responsibility.

Corresponding TA: Mingze Kong (mingzekong1@link.cuhk.edu.cn).

Problem 1 (10 pts)

Find the LU decomposition of the following matrix. Ensure you include all intermediate steps, including but not limited to row operations, that lead to the final LU factorization. Providing only the final answer without detailing the process will result in partial credit.

$$A = \begin{bmatrix} 4 & -20 & -12 \\ -8 & 45 & 44 \\ 20 & -105 & -80 \end{bmatrix}$$

Problem 2 (12 pts total, 4 pts each)

Use the matrix inverse to solve the following three systems of equations. More specifically, you shall first compute the inverse of the coefficient matrix (which is the same for all three systems), and then apply the formula of the solution for the three systems respectively.

$$(a) \begin{cases} 2x_1 + 3x_2 - x_3 &= 0 \\ x_1 + x_2 + x_3 &= 0 \\ 3x_1 - x_2 - x_3 &= 1 \end{cases}$$

$$(b) \begin{cases} 2x_1 + 3x_2 - x_3 &= 1 \\ x_1 + x_2 + x_3 &= 0 \\ 3x_1 - x_2 - x_3 &= -1 \end{cases}$$

$$(c) \begin{cases} 2x_1 + 3x_2 - x_3 &= -1 \\ x_1 + x_2 + x_3 &= 2 \\ 3x_1 - x_2 - x_3 &= 0 \end{cases}$$

Problem 3 (20 pts total, 4 pts each)

Determine whether the following statements are **True** or **False**. Justify your answers.

- (a) If A and B are invertible matrices of the same size, then AB and BA are both invertible.
- (b) Any elementary matrix is a triangular matrix.
- (c) The inverse of the elementary matrix $E = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ is $E^{-1} = \begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}$.
- (d) If an $n \times n$ matrix A is not invertible, then there are less than n pivots in Gauss-Jordan elimination for solving $Ax = 0$.
- (e) If an n -by- n matrix A that is not invertible, then the linear system $Ax = 0$ has more than one solution.

Remark: For judgment questions, you are suggested to write the solution as follows: "(a) True. Reason: ..." or "(a) False. Counter-example: ..."

Problem 4 (15 pts total, 5 pts each)

Solve the following problems and justify your answers:

- (a) Let $A = \begin{bmatrix} a & 0 \\ c & b \end{bmatrix}$. Prove that A is invertible if and only if $a \neq 0$ and $b \neq 0$.
- (b) Give an example of 2×2 invertible matrices A and B such that $A + B$ is not invertible.
- (c) Determine the relationship among a, b , and e that makes the following matrix invertible. Explain your answer.

$$A = \begin{bmatrix} 1 & 1 & a \\ 2 & 3 & b \\ 1 & 2 & e \end{bmatrix}$$

Problem 5 (14 pts total, 7 pts each)

Solve the following proof problems:

- (a) Suppose $A, M \in \mathbb{R}^{n \times n}$ are two square matrices, and $B, C \in \mathbb{R}^{n \times n}$ are invertible matrices such that $BA = CM$. Prove that A is invertible if and only if M is invertible.
- (b) Suppose $A \in \mathbb{R}^{n \times n}$ is a square matrix. Suppose there exists a positive integer k such that $A^k = \mathbf{0}$ (where $\mathbf{0}$ is the zero matrix). Let I be the identity matrix of order n .

Prove that: The matrix $I - A$ is invertible, and its inverse is given by:

$$(I - A)^{-1} = I + A + A^2 + \cdots + A^{k-1}$$

Problem 6 (13 pts total)

Let L and L_1 be $n \times n$ invertible lower triangular matrices, and let U and U_1 be $n \times n$ invertible upper triangular matrices. Suppose that $LU = L_1U_1$. Prove that there exists an invertible diagonal matrix D such that

$$L_1 = LD \quad \text{and} \quad U_1 = D^{-1}U.$$

Problem 7 (16 pts total, 4 pts each)

Determine whether each of the following sets is a **subspace** of the given vector space (with the usual addition and scalar multiplication). Briefly justify from the three conditions (contains zero, closed under addition, closed under scalar multiplication).

1. $W_1 = \{A \in \mathbb{R}^{2 \times 2} \mid A \text{ is symmetric}\} \subseteq \mathbb{R}^{2 \times 2}$ (i.e., $A^T = A$)
2. $W_2 = \{p(x) \in \mathcal{P}_2 \mid p(1) = 0\} \subseteq \mathcal{P}_2$, where $\mathcal{P}_2$ is the set of all polynomials of degree at most 2 with real coefficients.
3. $W_3 = \{p(x) \in \mathcal{P}_4 \mid p(x) = a_0 + a_2x^2 + a_4x^4 \text{ for some } a_0, a_2, a_4 \in \mathbb{R}\} \subseteq \mathcal{P}_4$, i.e., the set of polynomials of degree at most 4 that contain only even-degree terms.
4. $W_4 = \{(x, y) \in \mathbb{R}^2 \mid xy = 0\} \subseteq \mathbb{R}^2$